Roll No. | | | | | | | |

END SEMESTER EXAMINATION

Name of the Course: B.Tech Petroleum Engineering — *Semester: III*

Name of the Course: Thermodynamics — *Course Code*: TPE-301

Time: Three Hours — MM: 100

Note:

i. All questions are compulsory.
ii. Answer any two sub questions in each main question.
iii. Total marks for each main question is **twenty**
iv. Each Question carries **10 marks**

Q1		COs
a.	Why van der waals equation to be studied? Derive the constant values of 'a' and 'b' in terms of critical temperature and pressure	CO1, CO2
b.	A reversible Heat engine receives heat from a high temperature reservoir at T_1 K and rejects heat to a low temperature sink at 600 ^{0}C. a second Reversible engine receives the heat rejected by the 1st engine at 60 ^{0}C. Calculate the temperature T_1, for equal thermal efficiencies of the two engines	CO1, CO2
c.	Explain the three specifications for the quantitative definition of entropy? Define Clausius inequality and hence derive: $\oint \frac{dQ}{T} \leq 0$ for the Clausius inequality.	CO3
Q2		
a.	Illustrate Adiabatic process and hence derive an expression for the work done under adiabatically condition of the system. Specify the symbol used.	CO3, CO4
b.	A reversible Heat engine receives heat from a high temperature reservoir at T_1 K and rejects heat to a low temperature sink at 600 ^{0}C. a second Reversible engine receives the heat rejected by the 1st engine at 60 ^{0}C. Calculate the temperature T_1, for equal thermal efficiencies of the two engines	CO1, CO4
c.	Establish the relationship between COP of heat pump and COP of refrigerator, Hence solve: A house requires 2 x 105 kJ/h for heating in winter. Heat pump is used absorb heat from cold air outside in winter and send heat to the house. Work required to operate the heat pump is 3 x 104 kJ/h. Determine: i. Heat absorbed from outside ii. Coefficient of Performance [COP]	CO1, CO5
Q3		
a.	What do you mean by exact differential equation? Derive the all four Maxwell relation using fundamental property relations with the exact differential equation method? Specify the symbol used in deriving the relations.	CO1

b.	Establish the entropy-heat capacity relationships and hence: Calculate the entropy change when 1kmol of ideal gas at 300 K and 10 bar expands through a throttle to a pressure of 1 bar, both pressure being maintained during the process by suitable means.	CO1, CO2
c.	Discuss the effect of temperature and pressure on fugacity? Derive the expression for the fugacity of pure gases using compressibility factor?	CO2, CO5
Q4		
a.	Explain Gibbs free energy and hence Derive the modified equations for Internal energy and enthalpy by using the energy and Maxwell's relations? Specify the symbol used.	CO1, CO5
b.	Define activity? Deduce an expression for the activity coefficient by the effect of pressure. Determine the activity of a solid magnesium (MW = 24.32) at 300 K and 10 bar if the reference state is 300 K and 1 bar. The density of magnesium at 300 K is 1.745×10^3 kg/m^3 and is assumed constant over this pressure range	CO4, CO5
c.	Explain fugacity and fugacity coefficient? Derive the expression for the determination of fugacity of pure gases by using compressibility factor Z.	CO1, CO6
Q5		
a.	Derive an expression for the chemical potential with the help of Gibbs free energy. Mention the equation when it is reduced for isothermal process	CO4, CO6
b.	Discuss "The Linde process" for gas-liquefaction along with the component construction arrangements (diagram) of equipments used for the process.	CO4, CO5
c.	What do you mean by refrigerant? To design the refrigeration process the type of refrigerant is very important. What are the characteristics parameter on which the choice of refrigerant for the refrigeration process is to be considered? Discuss in details.	CO2, CO6